PORTFOLIO PROJECT

Landing Page Conversion Optimization using A/B Testing

A statistical evaluation of e-commerce user behavior and conversion uplift through controlled experimentation.

Data Analyst • Python • SciPy • Statsmodels

Problem Statement

The Challenge

An e-commerce company launched a redesigned landing page to boost user conversions, but needed hard evidence before a full rollout.

Business Goal

Determine if the new landing page converts users significantly better than the existing version to maximize ROI.

The Objective

Evaluate the statistical significance of conversion rate differences between the Control (old) and Treatment (new) groups.

Project Overview

Dataset Metadata

Attribute	Value
Total Records	294,478
Exp. Groups	Control / Treatment
Traffic Split	~50 / 50
Outcome	Binary (Converted: 1, No: 0)

Key Performance Indicators

- **Conversion Rate:** Percentage of users who converted.
- **Z-Statistic:** Standardized test score.
- **P-Value:** Significance threshold (alpha = 0.05).
- **Confidence Interval:** Range of plausible rate differences.

Business Understanding

Null Hypothesis (H_0)

The new landing page does not improve the conversion rate. Any observed difference is due to random chance.

Alternative Hypothesis (H_1)

The new landing page produces a statistically higher conversion rate. The improvement is real and repeatable.

"A/B testing isolates the effect of a design change from natural traffic variance."

Data Cleaning & Preparation

1. **Missing Value Detection:** df.isnull().sum() verified 0 missing records.

2. **Duplicate Control:** Cleansed double exposure tokens to maintain session integrity.

3. **Integrity Filtering:** Excluded group mismatches (Control with new page, Treatment with old page).

4. **Feature Engineering:** Segmented time arrays into discrete hour, day, and weekday matrices.

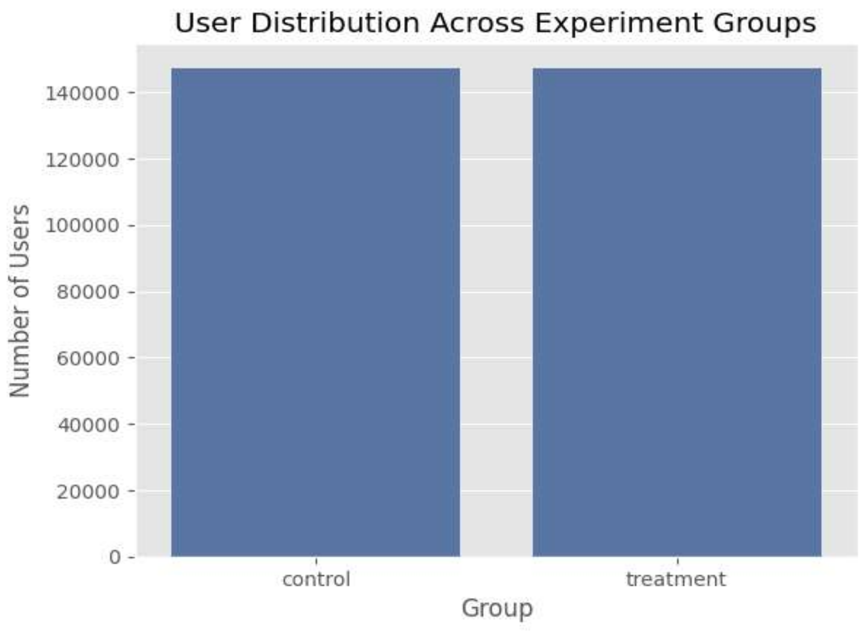
User Distribution Analysis

Balanced Sample Sizing

Random allocation achieved a highly dependable split across experimental variations:

Treatment (New Page): 147,276 users (50.01%)

Control (Old Page): 147,202 users (49.99%)



Conversion Performance Insights

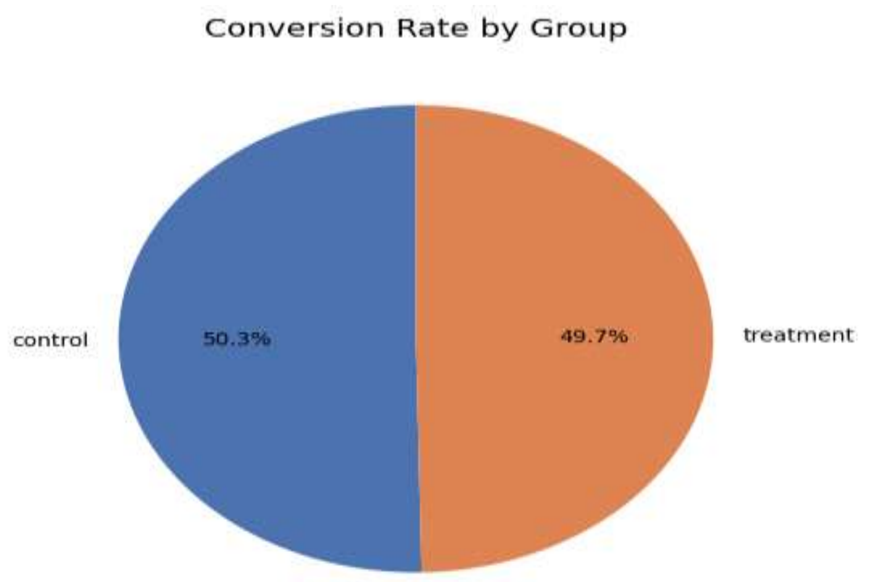
Observed Discrepancy

-0.15 pp

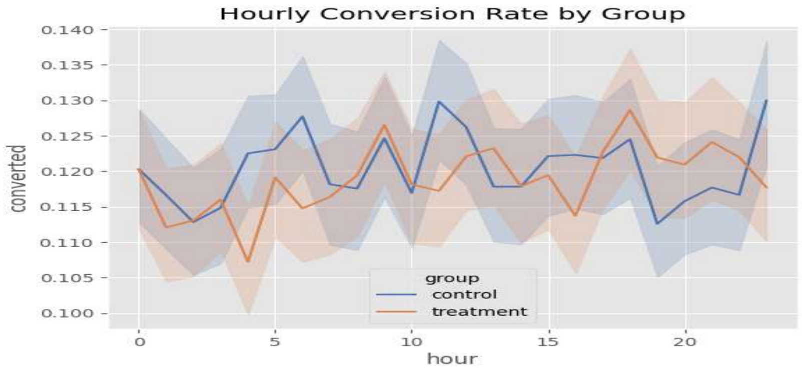
The treatment variant showed a nominal deficit compared to baseline metrics.

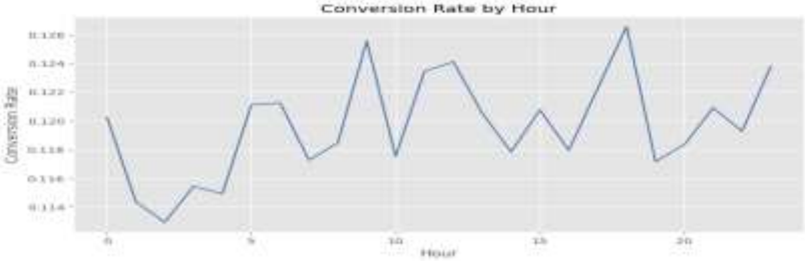
Baseline Comparison

Visual summaries demonstrate baseline parity. The graphic confirms no direct variant advantage.



Hourly Behavioral Patterns





Temporal Consistency

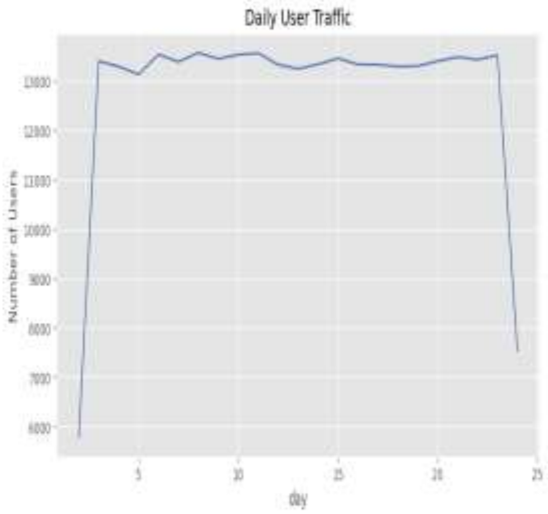
Conversion metrics track in parallel intervals throughout the 24-hour cycle. The structural redesign didn't shift chronological friction parameters.

Daylight Stability

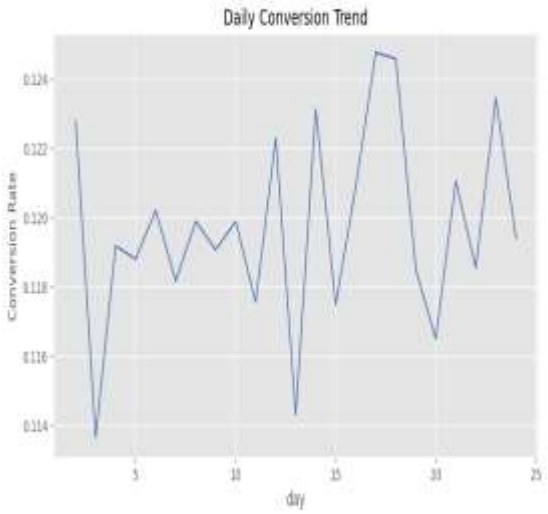
Variance metrics stay aligned through core usage windows, indicating uniform test conditions.

Traffic Volume & Trend Analysis

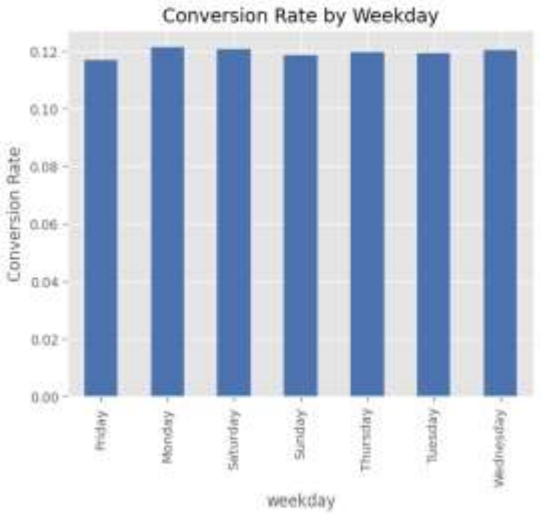
Daily User Traffic



Daily Conversion Trend



Conversion Rate by Weekday



Operational Stability: Volumes and weekday distributions verify reliable experiment continuity.

Statistical Hypothesis Testing

Calculated P-Value

0.905

ALPHA CRITERIA: 0.05

Retain Null Hypothesis (H_0)

The determined probability scale heavily exceeds mathematical boundaries. We cannot conclude that the new page induces systemic optimization.

Z-Score Value: -1.311

Statistical Inference: Insignificant Variance

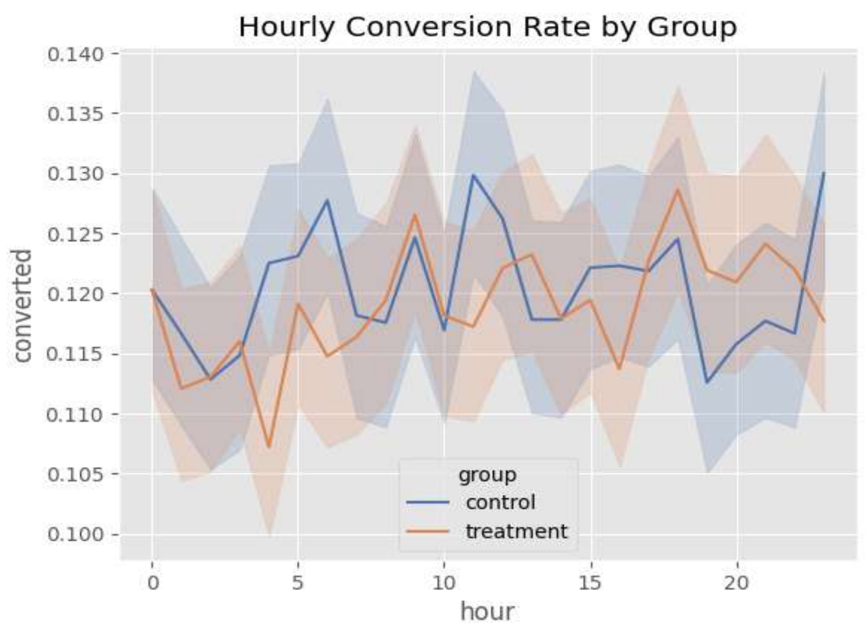
Confidence Intervals

Proportion Disparity Ranges

The 95% Confidence Interval boundaries measuring rate variance register as follows:

[-0.38%, +0.08%]

Since the mathematical limits span directly across zero, the measured variance fails evaluation checks and represents standard system noise.



Business Recommendations

⚠ DECISION: Do Not Deploy the New Landing Page at This Time

Retain Current Asset

Preserve current landing design to secure conversion baselines without technical implementation overheads.

Audience Dissection

Isolate user records by system variables or geographic tags to inspect hidden core performance pockets.

Alternative Vectors

Formulate a distinct design variation focusing heavily on isolated interaction friction points.

Thank You

Statistical evidence is the foundation of effective e-commerce strategy.

Data Analyst Portfolio Project | Python | Statistics | Experimentation | Business Analytics